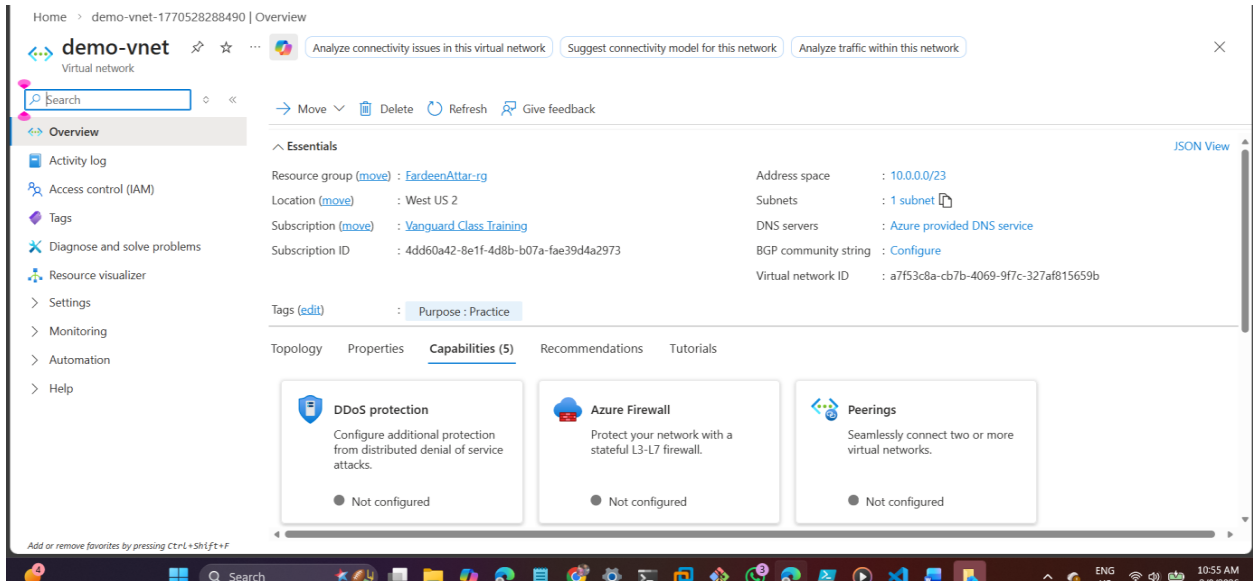


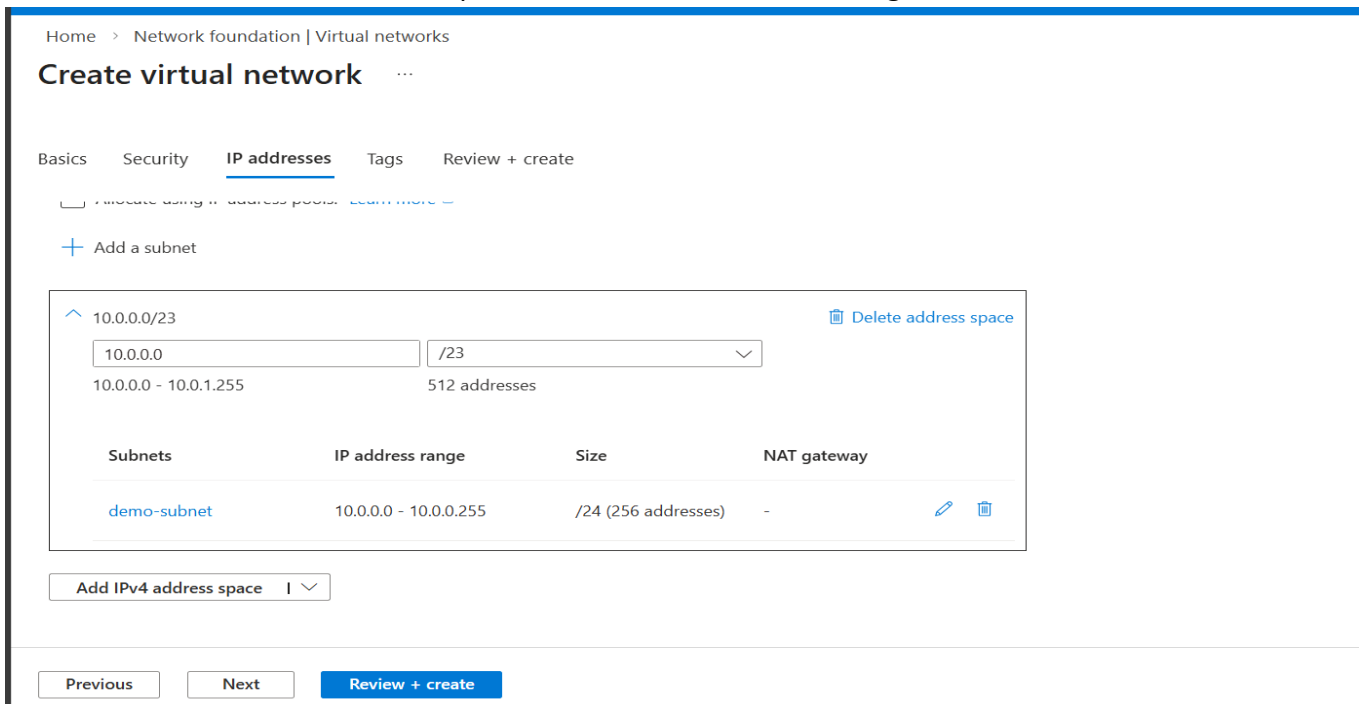
Day 5 – Assessment: The Infrastructure Challenge

Task 1: Networking Setup

- Create a **Virtual Network (VNet)**



- Define a **Subnet** within that VNet.
- **Goal:** Ensure the address space is sufficient for future scaling.



Home > Network foundation | Virtual networks

Create virtual network ...

BasicsSecurityIP addressesTagsReview + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
Purpose	Practice	All resources selected
		All resources selected

Home > demo-vnet-1770528288490 | Overview

demo-vnet

Analyze connectivity issues in this virtual networkSuggest connectivity model for this networkAnalyze traffic within this network

Virtual network

Search

MoveDeleteRefreshGive feedback

OverviewActivity logAccess control (IAM)TagsDiagnose and solve problemsResource visualizerSettingsMonitoringAutomationHelp

Essentials

Resource group (move) : FardeenAttar-rg

Location (move) : West US 2

Subscription (move) : Vanguard Class Training

Subscription ID : 4dd60a42-8e1f-4d8b-b07a-fae39d4a2973

Address space : 10.0.0.0/23

Subnets : 1 subnet

DNS servers : Azure provided DNS service

BGP community string : Configure

Virtual network ID : a7f53c8a-cb7b-4069-9f7c-327af815659b

Tags (edit) : Purpose: Practice

TopologyPropertiesCapabilities (5)RecommendationsTutorials

DDoS protection

Configure additional protection from distributed denial of service attacks.

Not configured

Azure Firewall

Protect your network with a stateful L3-L7 firewall.

Not configured

Peerings

Seamlessly connect two or more virtual networks.

Not configured

vnet is ready

Task 2: Security & Entry Points

- Create a Network Security Group (NSG).

The screenshot shows the Azure portal interface for a Network Security Group (NSG) named 'demo'. The left sidebar contains navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings (Inbound security rules, Outbound security rules, Network interfaces, Subnets, Properties, Locks), Monitoring, and Automation. The main content area displays the 'Overview' tab with the following details:

- Resource group: FardeenAttar-rg
- Location: West US 2
- Subscription: Vanguard_Class_Training
- Subscription ID: 4dd60a42-8e1f-4d8b-b07a-fae39d4a2973
- Tags: Purpose: Practice

Below the essentials, there are filters for Port, Protocol, Source, Destination, and Action, all set to 'all'. A table lists the Inbound Security Rules:

Priority	Name	Port	Protocol	Source	Destination	Action
100	RDP	3389	TCP	106.221.193.208	Any	Allow
110	SSH	22	TCP	106.221.193.208	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

- Configure a rule that allows you to remotely manage the VM.

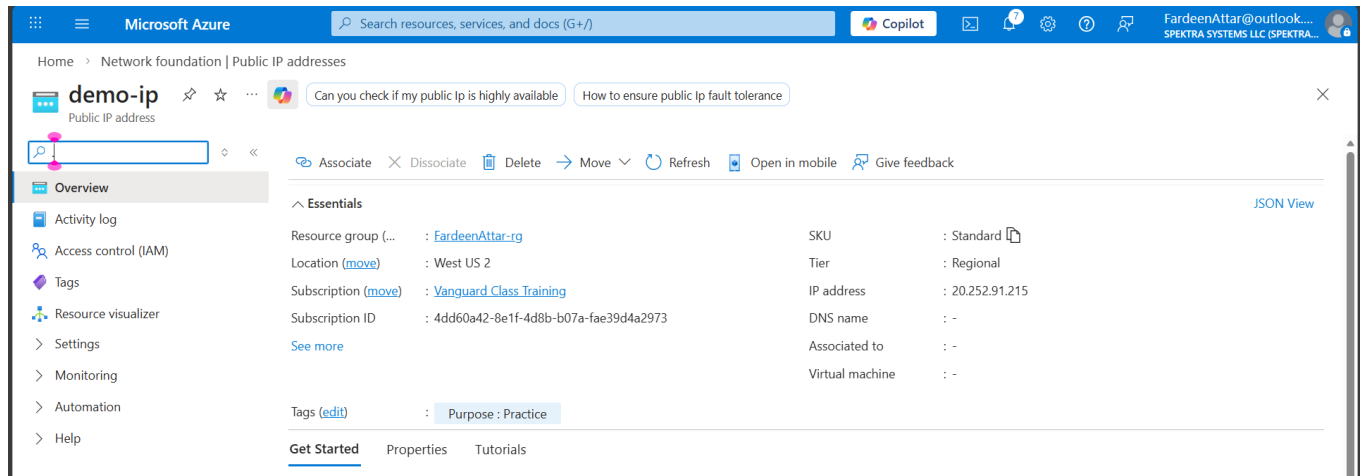
The screenshot shows the 'demo | Inbound security rules' configuration page in the Azure portal. The left sidebar is the same as the previous screenshot. The main content area displays the 'demo | Inbound security rules' tab with the following details:

- Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority. [Learn more](#)

Below the text, there are filters for Port, Protocol, Source, Destination, and Action, all set to 'all'. A table lists the Inbound Security Rules:

Priority	Name	Port	Protocol	Source	Destination	Action
☐ 100	RDP	3389	TCP	103.172.208.161	Any	Allow
☐ 110	SSH	22	TCP	106.221.193.208	Any	Allow
☐ 65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
☐ 65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	Allow
☐ 65500	DenyAllInBound	Any	Any	Any	Any	Deny

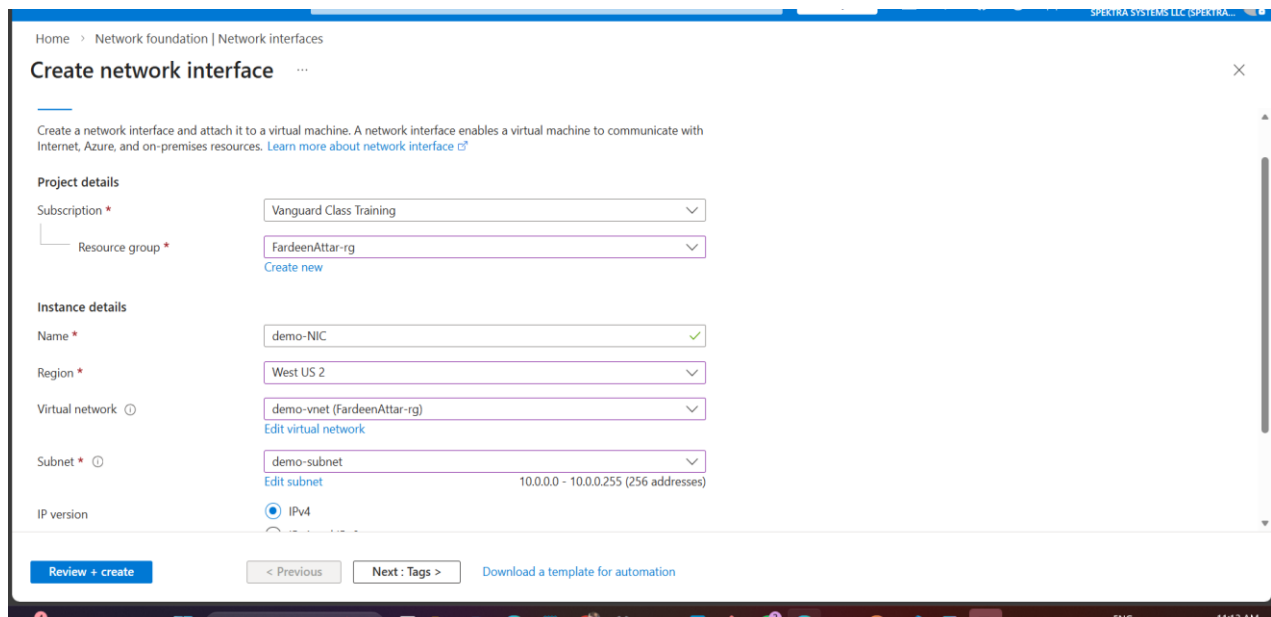
- **Create a Public IP Address to allow external connectivity.**



Public IP Address Created

Task 3: The Connection Layer

- Create a **Network Interface (NIC)**.
- Manually associate the VNet, Subnet, NSG, and Public IP you created in the previous steps.



Home > CreateNetworkInterfaceBladeV2-20260208111241 | Overview

demo-NIC Network Interface

Why can't this network interface connect to its VM? Analyze connectivity issues with this NIC

Search

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (...): [FardeenAttar-rg](#)

Location ([move](#)): West US 2

Subscription ([move](#)): [Vanguard Class Training](#)

Subscription ID: 4dd60a42-8e1f-4d8b-b07a-fae39d4a2973

Accelerated network...: Disabled

Virtual network/subnet: [demo-vnet/demo-subnet](#)

Tags ([edit](#)): Purpose: Practice

Private IPv4 address: 10.0.0.4

Public IPv4 address: -

Private IPv6 address: -

Public IPv6 address: -

Attached to: -

Type: Regular

Get started Properties

Created NIC

Associated Public IP Address to NIC

Home > Network foundation | Public IP addresses

demo-ip Public IP address

Can you check if my public ip is highly available? How to ensure public ip fault tolerance

Search

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Monitoring

Automation

Help

Associate Dissociate Delete Move Refresh Open in mobile

Essentials

Resource group (...): [FardeenAttar-rg](#)

Location ([move](#)): West US 2

Subscription ([move](#)): [Vanguard Class Training](#)

Subscription ID: 4dd60a42-8e1f-4d8b-b07a-fae39d4a2973

See more

Tags ([edit](#)): Purpose: Practice

Get Started Properties Tutorials

Use public IP addresses for public IP addresses

Associate and configure public IP address

Associate to a resource Configure a public IP

Associate public IP address

Choose the resource to which you want to associate this public IP address.

Resource type: Network interface

Network interface *: demo-NIC resource group: FardeenAttar-rg

OK Give feedback

Home > Network foundation | Public IP addresses

demo-ip Public IP address

Can you check if my public ip is highly available? How to ensure public ip fault tolerance

Search

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Monitoring

Automation

Help

Associate Dissociate Delete Move Refresh Open in mobile Give feedback

Essentials

Resource group (...): [FardeenAttar-rg](#)

Location ([move](#)): West US 2

Subscription ([move](#)): [Vanguard Class Training](#)

Subscription ID: 4dd60a42-8e1f-4d8b-b07a-fae39d4a2973

See more

Tags ([edit](#)): Purpose: Practice

Get Started Properties Tutorials

SKU: Standard

Tier: Regional

IP address: 20.252.91.215

DNS name: -

Associated to: [demo-NIC](#)

Virtual machine: -

Saved network interface

Successfully saved IP address changes for network interface 'demo-NIC'.

Task 4: Storage Allocation

- Create a **Managed Data Disk**.
- **Constraint:** This disk must be physically located to ensure zero-latency attachment to your compute resource.

The screenshot displays the Azure portal interface for a Managed Data Disk named 'demo-disk'. The breadcrumb navigation shows 'Home > Storage center | Azure Disks'. The disk is located in the 'FardeenAttar-rg' resource group. The 'Overview' tab is selected, showing various properties and a left-hand navigation menu. The 'Essentials' section lists key attributes: Resource group (FardeenAttar-rg), Disk state (Unattached), Location (West US 2), Subscription (Vanguard Class Training), Subscription ID (4dd60a42-8e1f-4d8b-b07a-fae39d4a2973), Time created (2/8/2026, 11:27:13 AM), and Tags (Purpose: Practice). The 'Properties' tab is also visible, showing details like Operating system type (Empty), Create option (Empty), and VM generation (1.0). The 'Size' tab shows the disk size (128 GiB), Storage type (Standard SSD LRS), and IOPS (500).

Home > Storage center | Azure Disks

demo-disk
Disk

Show protection details for this disk

Search

+ Create VM + Create VM image version + Create snapshot Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move) : FardeenAttar-rg

Disk state : Unattached

Last ownership updat... : -

Location : West US 2

Subscription (move) : Vanguard Class Training

Subscription ID : 4dd60a42-8e1f-4d8b-b07a-fae39d4a2973

Time created : 2/8/2026, 11:27:13 AM

Tags (edit) : Purpose : Practice

Disk size : 128 GiB

Storage type : Standard SSD LRS

Managed by : -

Operating system : -

Max shares : 0

Availability zone : 1

Security type : Standard

Get started Properties Monitoring

Disk

Operating system type : -

Create option : Empty

VM generation : 1.0

Size

Size : 128 GiB

Storage type : Standard SSD LRS

IOPS : 500

Created data disk in the **same AZ** where vm is going to create to ensure low latency .

Task 5: Final Assembly (Compute)

- Deploy a **Virtual Machine**.
- **Requirement:** Do not create new networking or storage during this step. You must attach the NIC and the Data Disk you prepared in Tasks 3 and 4.

Home

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Basics Disks **Networking** Management Monitoring Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network [Edit virtual network](#)

Subnet [Edit subnet](#) 10.0.0.0 - 10.0.0.255 (256 addresses)

Public IP [Create new](#)
Public IP addresses have a nominal charge. [Estimate price](#)

NIC network security group ☐ None ☒ Basic ☐ Advanced

< Previous Next: Management > **Review + create** [Give feedback](#)

- Attached the existing **vnet** and **Subnet** to my **demo-VM**

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a VM optimized for high availability Help me create a low cost VM Help me choose the right VM size for my workload

OS disk type [The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.](#)

Delete with VM ☒

Key management

Enable Ultra Disk compatibility ☐

Data disks for demo-VM

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM
0	demo-disk	128	Standard SSD LRS	Read/write	☐

[Create and attach a new disk](#) [Attach an existing disk](#)

< Previous Next: Networking > **Review + create** [Give feedback](#)

- Attached data disk to vm in same Region in same Zone to ensure low latency

VM IS READY

Home

demo-VM Virtual machine

Help me copy this VM in any region Manage this VM with Azure CLI

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Connect Start Restart Stop Hibernate Capture Delete Refresh Scale Open in mobile Feedback CLI / PS

Essentials

Resource group (move) : FardeenAttar-rg

Status : Running

Location : West US 2 (Zone 1)

Subscription (move) : Vanguard Class Training

Subscription ID : 4dd60a42-8e1f-4d8b-b07a-fae39d4a2973

Availability zone : 1

Operating system : Linux (ubuntu 24.04)

Size : Standard L2a0s v4 (2 vcpus, 16 GiB memory)

Primary NIC public IP : 20.252.91.215

Virtual network/subnet : demo-vnet/demo-subnet

DNS name : Not configured

Health state : -

Time created : 2/8/2026, 6:05 AM UTC

Tags (edit) : Purpose : Practice

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name : demo-VM

Networking

Public IP address : 20.252.91.215 (Network interface demo-NIC)

- Attached NSG to existing NIC

Home > Compute infrastructure | Virtual machines > demo-VM | Network settings > demo-NIC

demo-NIC | Network security group

Network interface

Search

Save Discard

Network security group : demo

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

IP configurations

DNS servers

Network security group

Properties

Locks

Monitoring

Automation

Admin

Home > Compute infrastructure | Virtual machines > demo-VM

demo-VM | Network settings

Virtual machine

What are the requirements for attaching and detaching network interfaces? How can I make this VM secure? +1

Search

List all my network interfaces for demo-VM What are the requirements for attaching or detaching a network interface?

Attach network interface Detach network interface View topology Troubleshoot Refresh Give feedback

Network interface / IP configuration

demo-NIC / ipv4config (primary)

demo-vm650_z1 (primary) / ipconfig1 (primary)

demo-NIC / ipv4config (primary)

Virtual network / sub... : demo-vnet / demo-subnet

Public IP address : 20.252.91.215

Private IP address : 10.0.0.4

Admin security rules : 0 (Configure)

Load balancers : 0 (Configure)

Application security ... : 0 (Configure)

Network security gro... : -

Accelerated network... : Disabled

Effective security rules : 0

Network security group demo (attached to networkInterface: demo-NIC)

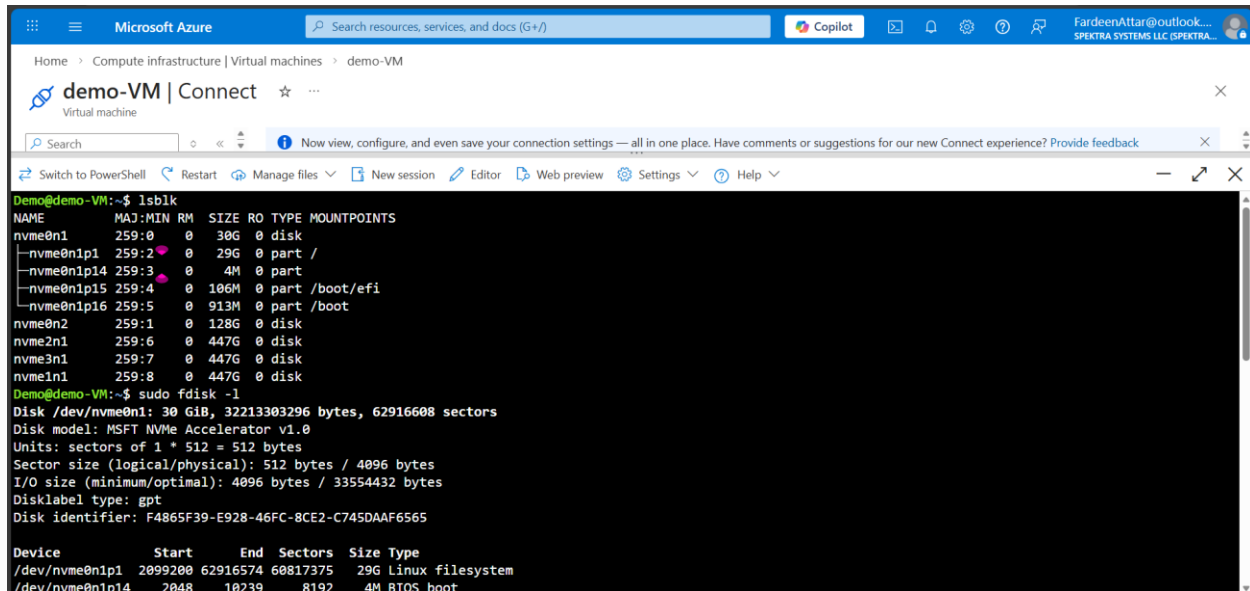
Impacts 0 subnets, 1 network interfaces

Create port rule

- Stopped VM and Attached existing Network Interface (NIC)

Task 6: Validation

- Log in to the machine and verify that the extra data disk is visible and ready for use



```
Demo@demo-VM:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1     259:0    0   30G  0 disk 
├─nvme0n1p1 259:2    0   29G  0 part /
├─nvme0n1p14 259:3    0    4M  0 part 
├─nvme0n1p15 259:4    0  106M  0 part /boot/efi
└─nvme0n1p16 259:5    0   913M  0 part /boot
nvme0n2     259:1    0  128G  0 disk 
nvme2n1     259:6    0  447G  0 disk 
nvme3n1     259:7    0  447G  0 disk 
nvme1n1     259:8    0  447G  0 disk 

Demo@demo-VM:~$ sudo fdisk -l
Disk /dev/nvme0n1: 30 GiB, 32213303296 bytes, 62916608 sectors
Disk model: MSFT NVMe Accelerator v1.0
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 4096 bytes
I/O size (minimum/optimal): 4096 bytes / 33554432 bytes
Disklabel type: gpt
Disk identifier: F4865F39-E928-46FC-8CE2-C745DAAF6565

Device            Start      End  Sectors  Size Type
/dev/nvme0n1p1    2099200 62916574 60817375 29G Linux filesystem
/dev/nvme0n1p14    2048    10239    8192    4M BIOS boot
```

Topology

